

## lab2uint16 Function Test Results

### Test Case 1: Typical L\*a\*b\* Image

"Input type:" "constant"

"Output type:" "uint16"

"Output:"

(:,:,1)

32640 48960

16320 65280

(:,:,2)

32768 37888

27648 43008

(:,:,3)

32768 25088

40448 48128

### Test Case 2: Minimum Valid Values (uint8 Input)

"Input type:" "uint8"

"Output type:" "uint16"

"Output:"

(:,:,1)

0 0

0 0

(:,:,2)

0 0

0 0

(:,:,3)

0 0

0 0

### **Test Case 3: Maximum Valid Values (uint16 Input)**

**"Input type:"** "uint16"

**"Output type:"** "uint16"

**"Output:"**

(:,:,1)

65280 65280

65280 65280

(:,:,2)

65280 65280

65280 65280

(:,:,3)

65280 65280

65280 65280

### **Test Case 4: Single Pixel**

**"Input type:"** "constant"

**"Output type:"** "uint16"

**"Output:"**

(:,:,1)

32640

(:,:,2)

32768

(:,:,3)

32768

#### **Test Case 5: uint8 Input**

**"Input type:" "uint8"**

**"Output type:" "uint16"**

**"Output:"**

(:,:,1)

0 32768

65280 16384

(:,:,2)

32768 32768

32768 32768

(:,:,3)

32768 32768

32768 32768

#### **Test Case 6: uint16 Input**

**"Input type:" "uint16"**

**"Output type:" "uint16"**

**"Output:"**

(:,:,1)

0 1000

30000 65280

(:,:,2)

0 1000

30000 65280

(:,:,3)

0 1000

30000 65280

### **Test Case 7: Out-of-Range Values**

**"Input type:"** "constant"

**"Output type:"** "uint16"

**"Output:"**

(:,:,1)

0 65535

65535 0

(:,:,2)

0 65535

0 65535

(:,:,3)

0 65535

0 65535

### **Test Case 8: NaN Values**

**"Input type:"** "constant"

**"Output type:"** "uint16"

**"Output:"**

(:,:,1)

65535 32640

(:,:,2)

32768 32768

(:,:,3)

32768 32768

#### **Test Case 9: Invalid 2-D Input**

**"Output:"** "lab2uint16: LAB must be Mx3, MxNx3, or MxNx3xK size"

#### **Test Case 10: int16 Input**

**"Output:"** "lab2uint16: invalid class for LAB"